1. Creating and Renaming Files/Directories

a. Create a directory named test_dir using mkdir.

```
root@Master:/home/krishna# mkdir test_dir
root@Master:/home/krishna# ls
test_dir
```

b. Inside test_dir, create an empty file called example.txt.

```
root@Master:/home/krishna# cd test_dir/
root@Master:/home/krishna/test_dir# touch example.txt
root@Master:/home/krishna/test_dir# ls
example.txt
```

c. Rename example.txt to renamed_example.txt using mv

```
root@Master:/home/krishna/test_dir# mv example.txt renamed_example.txt
root@Master:/home/krishna/test_dir# ls
renamed_example.txt
```

A Sairam Kumar Pro

2. Viewing File Contents

a. Use cat to display the contents of /etc/passwd.

```
root@Master:/home/krishna/test_dir# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
krishna:x:1000:1000:,,,:/home/krishna:/bin/bash
```

b. Display only the first 5 lines of /etc/passwd using head.

```
root@Master:/home/krishna/test_dir# head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

c. Display only the last 5 lines of /etc/passwd using tail.

```
root@Master:/home/krishna/test_dir# tail - n 5 /etc/passwd
==> standard input <==
1
2
3
4
5
^Z
[1]+  Stopped                 tail - n 5 /etc/passwd
```

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
root@Master:/home/krishna/test_dir# grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

4. Zipping and Unzipping

a. Compress the test_dir directory into a file named test_dir.zip using zip.

```
root@Master:/home/krishna# apt install zip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  unzip
The following NEW packages will be installed:
  unzip zip
0 upgraded, 2 newly installed, 0 to remove and 215 not upgraded.
Need to get 176 kB/350 kB of archives.
After this operation, 933 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zip amd64 3.0-13ubuntu0.2 [176 kB]
Fetched 176 kB in 33s (5299 B/s)
Selecting previously unselected package unzip.
(Reading database ... 40794 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4.1_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4.1) ...
Selecting previously unselected package zip.
Preparing to unpack .../zip_3.0-13ubuntu0.2_amd64.deb ...
Unpacking zip (3.0-13ubuntu0.2) ...
Setting up unzip (6.0-28ubuntu4.1) ...
Setting up zip (3.0-13ubuntu0.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
root@Master:/home/krishna# zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
root@Master:/home/krishna# ls
test_dir  test_dir.zip
```

b. Unzip test_dir.zip into a new directory named unzipped_dir.

```
root@Master:/home/krishna# unzip test_dir.zip -d unzipped_dir
Archive:  test_dir.zip
   creating: unzipped_dir/test_dir/
 extracting: unzipped_dir/test_dir/renamed_example.txt
root@Master:/home/krishna# ls
test_dir  test_dir.zip  unzipped_dir
```

5. Downloading Files

Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

```
root@Master:/home/krishna# wget https://sample-files.com/downloads/documents/txt/simple.txt
--2026-05-01 01:18:18--  https://sample-files.com/downloads/documents/txt/simple.txt
Resolving sample-files.com (sample-files.com)... 15.235.219.232
Connecting to sample-files.com (sample-files.com)|15.235.219.232|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 225 [text/plain]
Saving to: 'simple.txt'

simple.txt                100%[===================================>]     225  --.-KB/s    in 0s

2026-05-01 01:18:19 (6.79 MB/s) - 'simple.txt' saved [225/225]

root@Master:/home/krishna# ls
simple.txt  test_dir  test_dir.zip  unzipped_dir
```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
root@Master:/home/krishna# touch secure.txt
root@Master:/home/krishna# chmod 444 secure.txt
root@Master:/home/krishna# ls -l secure.txt
-r--r--r-- 1 root root 0 May  1 02:01 secure.txt
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
root@Master:/home/krishna# export MY_VAR="Hello, Linux!"
root@Master:/home/krishna# echo MY_VAR
MY_VAR
root@Master:/home/krishna# echo $MY_VAR
Hello, Linux!
```

Github repo link: https://github.com/itsairamkumar/linux-basics-devops-practice.git